

Economy in Research Communication

Harsh Torane

15 September 2026

Abstract

Research papers take time to write, review, and read. Authors may therefore benefit from streamlining papers when material can be removed without weakening the communication of the research contribution. Lander and Parkin’s 1966 counterexample to Euler’s conjecture illustrates how a substantive mathematical result can be communicated in a single page without the structure of a conventional research article. This does not imply that shorter papers are inherently better. Rather, this paper proposes economy in research communication as a practical principle: when material can be removed without materially weakening the communication of a research contribution, it need not be retained simply because it is conventional. The principle concerns unnecessary material, not brevity for its own sake.

Introduction

Research papers require substantial effort from authors, reviewers, and readers. Contemporary scientific papers may contain lengthy main texts, figures, references, and supplementary material (“The Ever-Changing Communication of Scientific Discovery” 2025). Much of this material is necessary, but some may remain because of convention, repetition, or an assumption that greater length signals greater completeness.

This raises a practical question: **if a research contribution can be communicated clearly and rigorously with less material, why make the paper longer?**

Streamlining a paper may reduce the amount of material that authors, reviewers, and readers must process. Publication costs provide a secondary consideration. Some journals impose page, colour, submission, or other author-paid charges, although their prevalence and structure vary across disciplines and journals (Curb and Abramson 2012; Gray 2015). These costs do not establish that shorter papers are preferable, but they provide one practical reason to avoid unnecessary material where journal policies make additional length costly.

Lander and Parkin’s 1966 counterexample to Euler’s conjecture provides a useful illustration (Lander and Parkin 1966). Their communication reports a substantive mathematical result in a single page and does not follow the structure of a modern research article. The significance of the example is not simply its brevity. Rather, it illustrates that the amount and structure of communication required can depend on the nature of the result. A compact mathematical counterexample may require little exposition when the result itself is directly stated and verifiable.

This paper does not attempt to establish an optimal paper length or quantify the effects of manuscript compression. Instead, it proposes a simple evaluative principle for deciding whether individual components of a research paper justify their inclusion. The contribution is therefore conceptual: to distinguish economy in research communication from brevity as an end in itself.

Economy in Research Communication

The practical principle proposed here is:

When material can be removed without materially weakening the communication of the research, remove it.

Here, “materially” means sufficiently to impair a reasonable reader’s ability to understand, evaluate, reproduce, or appropriately interpret the research.

This principle does not require determining the shortest possible paper. It instead asks whether each part of a paper contributes sufficiently to its communication to justify its presence.

Economy should be distinguished from brevity and compression. Brevity concerns the amount of communication; compression concerns expressing necessary information more efficiently; economy concerns whether the material included in the communication justifies its cost in attention, effort, or other resources. A paper may therefore become more economical without becoming substantially shorter, for example by replacing a redundant explanation with a clearer one.

Some content is **necessary**, because removing it would impair understanding, evaluation, reproducibility, or interpretation. Some is **useful but non-essential**, because it improves explanation or accessibility but may be shortened. Other material is **redundant**, because removing it does not materially reduce the reader's ability to understand or evaluate the research.

The objective is to remove or compress material in the latter categories without sacrificing information necessary for the research to be properly understood and assessed.

Discussion

Economy in research communication can take ordinary forms: removing repetition, shortening unnecessary background, combining redundant explanations, reducing excessive sectioning, or moving genuinely non-essential material outside the main text. The appropriate degree of compression will vary substantially by field, research design, and intended readership.

A short paper is therefore not necessarily a better paper. A mathematical counterexample, a clinical study, and a systematic review may have very different requirements for sufficient communication. Complex methods, datasets, statistical procedures, limitations, negative results, and contextual information may require substantial space. Compression becomes counterproductive when it removes information necessary for interpretation or evaluation.

The potential benefits of avoiding unnecessary material are straightforward. Less redundant content may mean less time spent drafting and revising, less material for reviewers to process, and less material for readers to work through. However, the magnitude of these effects is an empirical question rather than a claim established by this paper.

The principle can therefore be understood as an economy principle for research communication. The objective is not to minimize length itself, but to avoid material that does not materially contribute to the communication of the research.

Limitations

Economy in research communication cannot be applied independently of disciplinary norms, research design, or readership. What may be unnecessary in one paper may be essential in another. Material that initially appears redundant may also serve important purposes for accessibility, transparency, replication, or interpretation.

The principle proposed here is consequently a heuristic rather than a universal rule. Its value depends on identifying material that can be removed or compressed without impairing the reader's ability to understand, evaluate, reproduce, or appropriately interpret the research.

Conclusion

Authors do not need to make papers short for the sake of being short. But when a paper can communicate its contribution clearly and rigorously with less unnecessary material, streamlining may benefit authors, reviewers, and readers.

The useful goal is therefore **economy in research communication**, rather than maximum length or brevity for its own sake. In practice, authors can ask of each part of a paper: **Does this materially help the reader understand, evaluate, reproduce, or appropriately interpret the research?** If not, it may not need to remain in the paper in its current form.

References

- Curb, L. A., and C. I. Abramson. 2012. “An Examination of Author-Paid Charges in Science Journals.” *Comprehensive Psychology* 1: Article 4. <https://doi.org/10.2466/01.17.CP.1.4>.
- Gray, A. 2015. “Considering Non-Open Access Publication Charges in the ‘Total Cost of Publication.’” *Publications* 3 (4): 248–62. <https://doi.org/10.3390/publications3040248>.
- Lander, L. J., and T. R. Parkin. 1966. “Counterexample to Euler’s Conjecture on Sums of Like Powers.” *Bulletin of the American Mathematical Society* 72 (6): 1079. <https://doi.org/10.1090/S0002-9904-1966-11654-3>.
- “The Ever-Changing Communication of Scientific Discovery.” 2025. *Nature Biomedical Engineering* 9: 1559–60. <https://doi.org/10.1038/s41551-025-01551-0>.